old device which was used in the 1980s?). And traffic was captured using a Wireshark system connected to one port on the Ethernet hub.

(d) This hit the jackpot (see Figure 1) revealing interesting malware on various computer systems. (Figure 1 shows only one malware, though there were many more, which I unfortunately could not capture.)

(e) Googling these links revealed a known malware (see Figure 2).

(f) Anti-virus and anti-spyware software were run on the systems to clear it of malware.

(g) Firewall black lists were updated to block these links.

Thus, removing malware solved the issue.

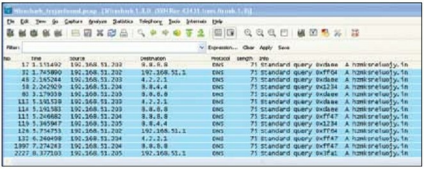

Figure 1: Malware in the capture

Scenario 3: A slow network and systems disconnect from it intermittently

The described problem was that any of the computer systems connected to the network would disconnect from the LAN intermittently. This would also lead to the complete network coming to a halt.

The in-house administrator had tried to diagnose the problem for several days. Discussions revealed several interesting facts. The installation had two 24-port Gigabit unmanaged Ethernet switches in two rooms connected by a crossover cable. LAN traffic was heavy with several large files being copied or moved between various systems every now and then. Each switch connected to 20 computer systems. The problem was first observed after half the older computers had been replaced with new systems. Network cabling and Ethernet switches were not changed or even touched during the upgrade. Operating system security updates were applied to the new computer systems. Rebooting the Ethernet switches would start the network, but the problem would reappear randomly.

The steps taken for troubleshooting were as follows:

- The first step was to check the operating systems – the problem was reported after the change (of computers). So, the latest drivers of the motherboard, particularly the Network Interface Card (NIC) were downloaded and applied. On various forums there were complaints about similar problems for a specific model of NICs (which were onboard for the new systems) supporting EEE (Energy Efficient Ethernet). If connected to switches not supporting EEE, random network disconnections were reported. So, EEE features were disabled – it was expected that the problem would be resolved, but that didn't happen.
- Anti-virus signatures of all the systems were updated, and the systems scanned for viruses. Some of the systems had viruses which were removed, but the problem reappeared.
- Switches were replaced using the hub, turn by turn. Wireshark was deployed to capture network traffic for analysis. To our utmost surprise, there was very heavy IPv6 traffic in the network measuring almost 77 per cent of the total traffic, even though the network was configured for IPv4. So, it appeared that the problem was located—protocol IPv6 was disabled from all the systems in the belief that there could be an unknown malware that was using IPv6 to spread! But, again, there was no luck here too. The problem persisted.

Wireshark Statistics menu: Here, let us take a small break to understand an interesting Wireshark menu – Protocol Hierarchy – which is available under Statistics. This could be used to arrive at high IPv6 bandwidth utilisation while

Figure 2: Malware identified (Screenshot from Sophosh.com website)